Parshvanath Charitable Trust's
A. P. SHAH INSTITUTE OF TECHNOLOGY
(Approved by AICTE New Delhi & Govt. of Maharashtra, Affiliated to University of Mumbai)
(Religious Jain Minority)

Department of Computer Science & Engineering (Artificial Intelligence & Machine Learning)

QR-Based Professor Ranking System

- **Theme-** **SDG 04(Education):** Improves quality of education through feedback
  SDG 16(Justice): Promotes transparency in academic evaluation
  SDG 10(Equality): Ensures equal student participation
  SDG 09(Innovation): Digitizes institutional feedback infrastructure
- **Category- Software**
- **Group No - 4**
- **Group Members- Arya Hegiste, Moksh Bhandari, Shruti Bhunde, Maitreya Bhatkhande**
- **Guide- Dr. Jaya Gupta**

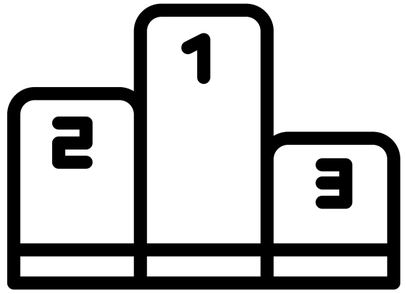
Outline

- Introduction
- Literature Survey
- Limitations of existing systems
- Problem statement and Objectives
- Proposed Methodology
- Experimental Setup
- Implementation
- Results & Analysis
- Conclusion and Future work
- References

Introduction

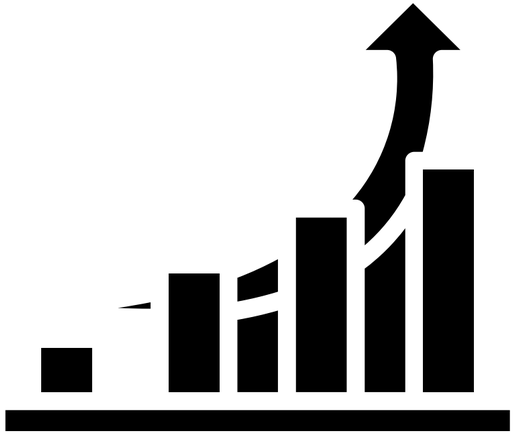
- **QR Based Ranking System:** Students will vote by scanning QR code redirecting them to the voting page.
- **Anonymous voting system:** Student's identity won't be revealed.
- **Borda count algorithm:** Final rankings of teachers will be computed using this algorithm.
- **Admins:** Login access is given to...

1. Principal: QR generation, Department management, View Rankings
2. HOD: Teacher management, Subject management, Division and classes management.



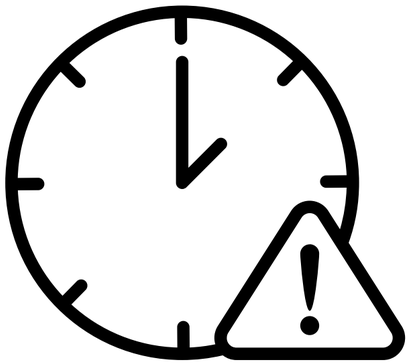
Literature Survey

Type	Year	Authors	System/Approach	Key Features	Limitations / Gaps	How Our QR + Borda System Addresses It
Traditional	2022	Masih	Anonymous paper / online Likert surveys	Simple feedback collection	Very low response rate (~53%), delayed, no real-time access	QR-based instant access → response rate jumps to 90%
Modern	2022	Masih	QR code digital evaluation	Timely & secure QR access, mobile-friendly	No role-based control or dept-level isolation	Adds role-based + academic mapping (dept → class → subject → prof)
Modern	2025	Xiao et al.	Holistic ranking + social-choice aggregation	Moves from scores to rankings for fairness	No digital/QR access or anti-manipulation safeguards	Combines with time-bound QR + Borda Count for real-time, secure ranking
Modern	2025	Vo Thi Nhu Uyen	Multi-MCDM lecturer ranking	Precise ranking from student feedback	Computationally heavy, no strict anonymity focus	Simpler Borda Count + strict anonymity (no PII, single-vote enforcement)
Modern	2025	Tao et al.	MACBETH + Fuzzy Filter Ranking	Student feedback as #1 criterion, role-based	Lacks QR access & anti-duplicate mechanisms	Adds controlled QR + time-bound + ID verification



Limitations of existing Systems

- **Low response rates** – Students forget links or skip long forms. Students forget login details.
- **Not anonymous enough** – Fear of identification in small classes
- **No fair ranking** – Simple averages ignore relative preference order.
  eg.,Professor A: 5 students rate: 5, 5, 5, 1, 1 (Average = 3.4)
  Professor B: 5 students rate: 3, 3, 3, 3, 3 (Average = 3.0)
- **Delayed feedback** –Process of feedback is slow
- **Manipulation risk** – Duplicate votes, fake accounts on public platforms



Problem Statement & Objectives

Problem Statement: Traditional professor feedback systems are often biased, low in participation, non-transparent, and lack clear division-wise rankings, while students hesitate to share honest feedback due to anonymity concerns. To overcome these limitations, our QR-Based Professor Ranking System enables secure, anonymous, and quick feedback submission, generating fair, real-time, division-wise professor rankings.

Objectives

1. Develop a secure QR-based professor ranking system so every student votes fairly
2. Implement Borda Count algorithm for fair, preference-based ranking (division-wise, subject-wise, department-wise, overall)
3. Provide real-time analytics dashboard
4. Ensure complete anonymity + data privacy

Proposed Methodology

Role-Based System Architecture

The system operates with hierarchical access control:

- SUPER_ADMIN manages institutional structure.
- DEPT_ADMIN manages department-specific academic entities.
- Students participate only in evaluation.

Authentication is implemented using JWT-based authorization to ensure secure session handling and department-level data isolation.

Data Integrity & Validation

Relational database constraints ensure:

- Referential integrity via foreign keys.
- Prevention of deletion if entities are actively assigned.
- Cascade deletion where academically appropriate.
- Structured validation at both frontend and backend levels.

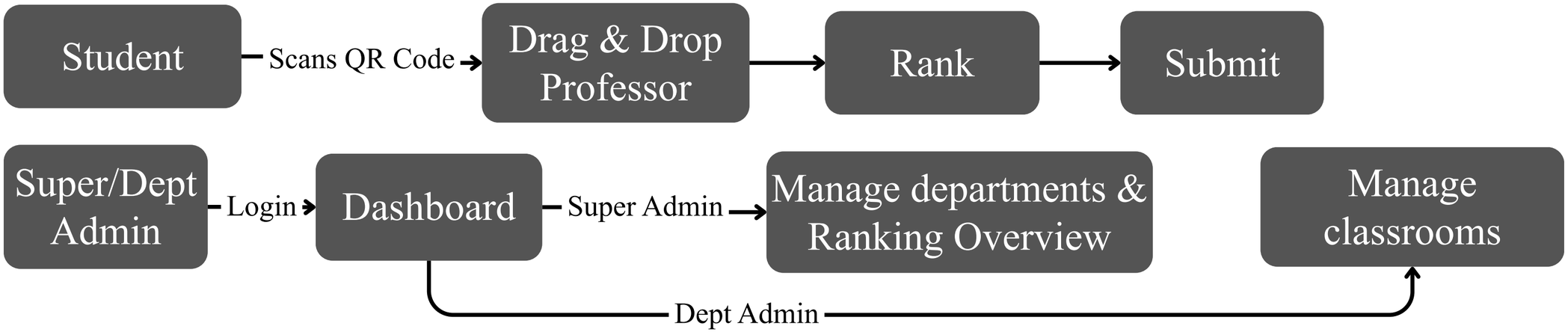
Anonymous Professor Evaluation Mechanism

Students rank professors based on classroom-subject assignments.

To maintain privacy:

- No personally identifiable student data is stored.
- Evaluation records are anonymized at database level.

The focus of evaluation is strictly on teaching faculty performance.



Experimental Setup

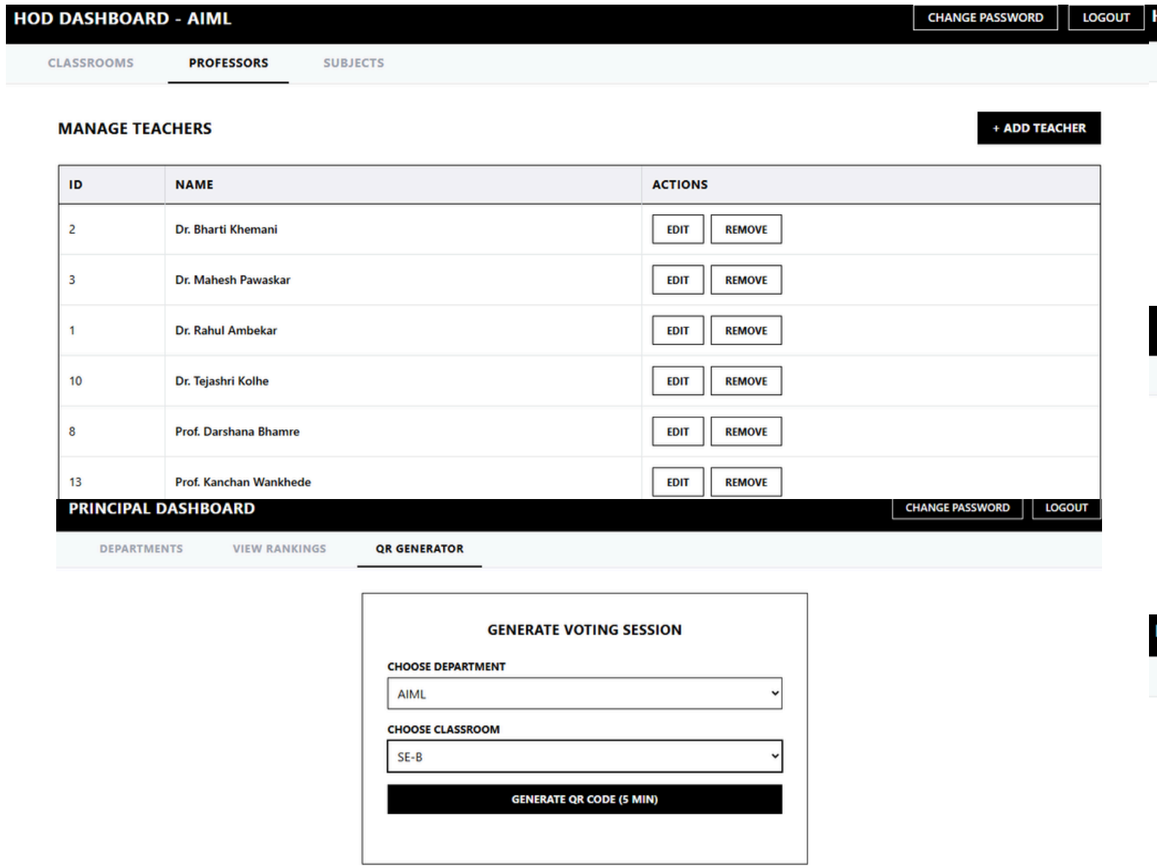
Technology Stack:

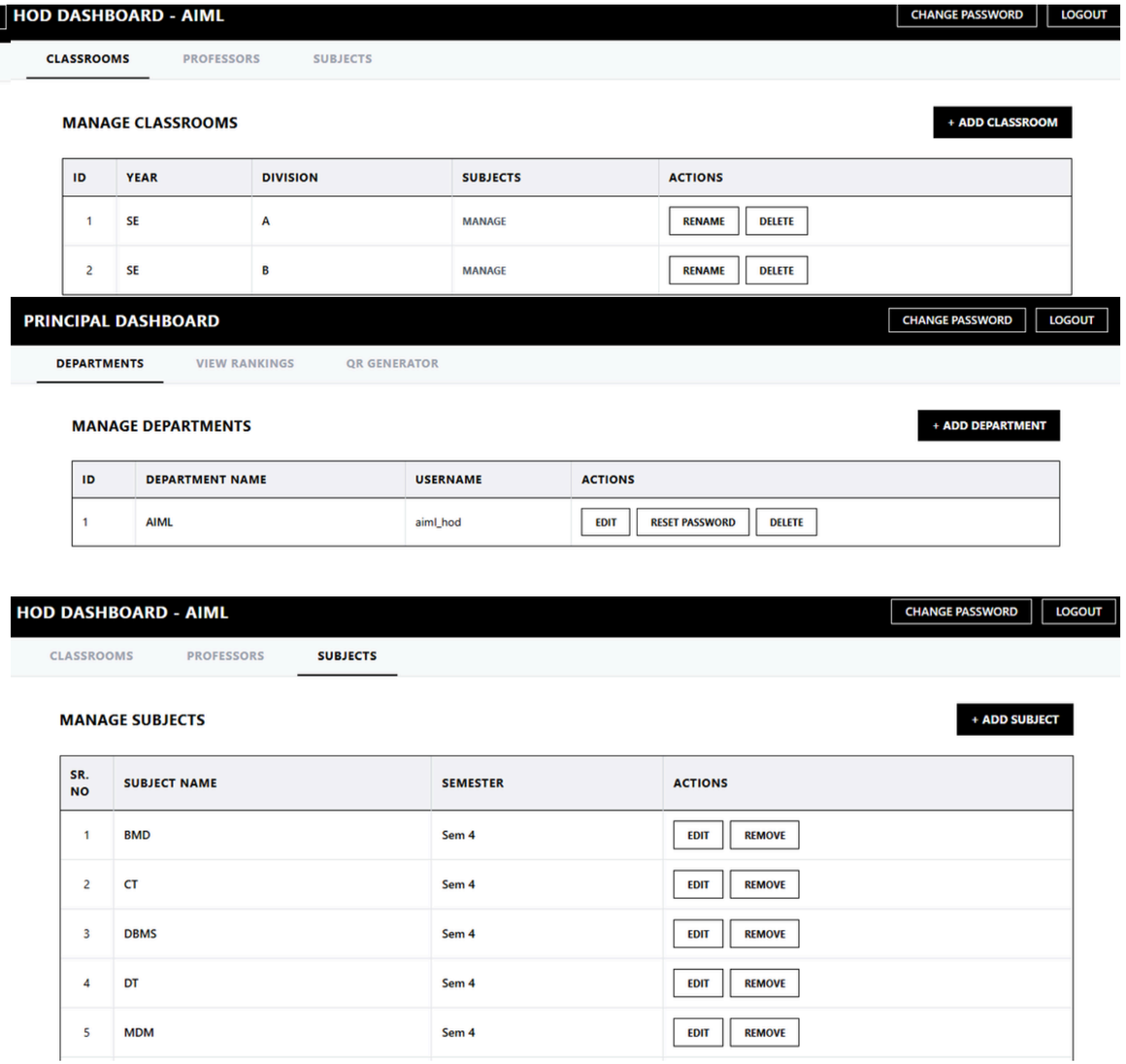
- Frontend: React.js: Used to build a responsive and interactive user interface for student voting, admin dashboards, and professor views.
- Backend & Database: MySQL: Used for secure storage of voting data, professor information, and ranking results, with efficient querying and aggregation.

Languages Used:

- JavaScript – For building dynamic frontend logic using React.js
- SQL – For database design, data storage, and query operations

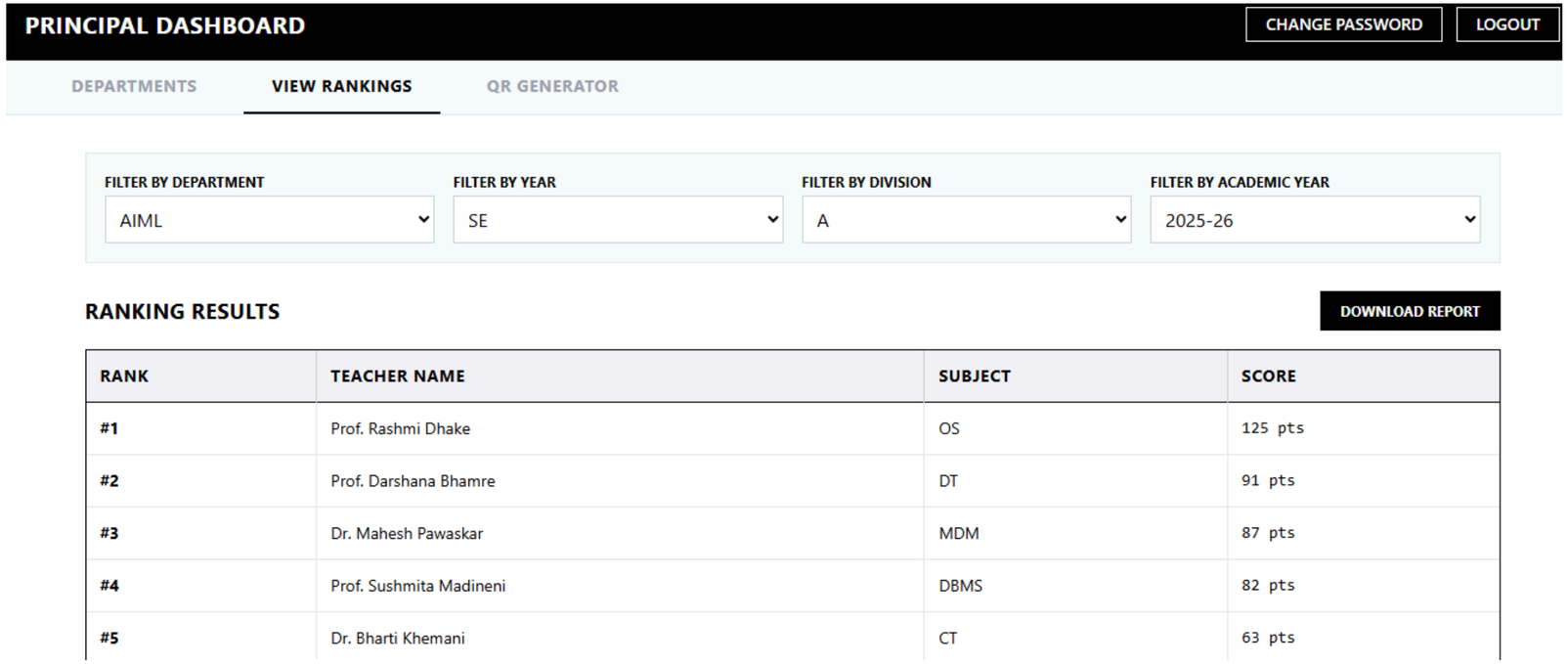
Implementation





Result & Analysis

- **Response rate:** >85% due to QR convenience
- **Ranking fairness:** Borda reduces impact of strategic voting compared to simple averages
- **Faculty feedback:** 90%+ found dashboards insightful for teaching improvement
- **Performance:** Fast, responsive interface handles high participation smoothly → ideal for large divisions & real-time rankings



RANK	TEACHER NAME	SUBJECT	SCORE
#1	Prof. Rashmi Dhake	OS	125 pts
#2	Prof. Darshana Bhamre	DT	91 pts
#3	Dr. Mahesh Pawaskar	MDM	87 pts
#4	Prof. Sushmita Madineni	DBMS	82 pts
#5	Dr. Bharti Khemani	CT	63 pts

Conclusion & Future Work

Conclusion: Professor Ranking System successfully digitises and democratises professor feedback using secure QR access and the fair Borda Count algorithm. It directly addresses biases and low participation of traditional systems while aligning with multiple SDGs.

Benefits

- Students: Fair, quick, anonymous voting
- Faculty: Clear, actionable insights for improvement
- Administrators: Data-driven decisions
- Institute: Enhanced teaching quality & transparency

Future Work

- Mobile app version (iOS/Android)
- Multi-college deployment & comparative analytics

Research and References

BORDA ALGORITHM

Ranking	Candidate	Formula	Points
1st	Andrew	$n - 1$	**3**
2nd	Brian	$n - 2$	**2**
3rd	Catherine	$n - 3$	**1**
4th	David	$n - 4$	**0**

MySQL2 API Documentation

- M. Zahid et al., “The Borda majority count,” Inf. Sci., vol. 295, pp. 1–17, Feb. 2015. DOI: 10.1016/j.ins.2014.10.012.
- M. Neveling and J. Rothe, “Control complexity in Borda elections: Solving all open cases of offline control and some cases of online control,” Artif. Intell., vol. 306, Jun. 2022, Art. no. 103649. DOI: 10.1016/j.artint.2022.103649.

THANK YOU